ADELE VALERIA

Brooklyn, New York City — hello.adelev@gmail.com — 6673511557 — linkedin.com/in/adelevaleria

SUMMARY

Computational biologist with 4 years of graduate research experience in computational immunogenomics and cancer neoantigen prediction. Experienced in applying computational methods, bioinformatics, and data science to biomedical research, with broader interests in technical education, technical customer-facing roles, and responsible AI.

EDUCATION

Johns Hopkins University August 2026
Master of Science in Engineering in Computer Science *Baltimore, United States*

Zhejiang University-University of Edinburgh Institute May 2022
Bachelor of Science in Biomedical Informatics *Haining, China*

Strathfield College May 2018
Advanced Diploma of IT Project Management and Diploma of IT *Sydney, Australia*

RESEARCH EXPERIENCE

HERV-Derived Neoantigen Pipeline Design 01/2026-03/2026
Dr. Valsamo Anagnostou at Johns Hopkins Medicine *Baltimore*

· Reviewed computational approaches for identifying neoantigen candidates derived from human endogenous retroviruses (HERVs) and conceptualized a preliminary pipeline.

Advancing Cancer Neoantigen Prediction 02/2024-12/2025
Dr. Rachel Karchin at Johns Hopkins University *Baltimore*

· Evaluated MHC (major histocompatibility complex) representation strategies for cancer neoantigen prediction, resulting in a first-author preprint on **bioRxiv:**: doi.org/10.64898/2025.12.09.693250
· Evaluated gene embedding approaches for integrating RNA-seq information into the open-source BigMHC deep learning framework for neoantigen prediction.
· Developed protein representations using ESM-2 and graph-based Node2Vec.
· Managed model training on GPU-based HPC infrastructure using Slurm.
· Represented the lab at the 2025 Nextflow/nf-core Hackathon in Boston, contributing to collaborative bioinformatics workflow development.
· Tools and techniques: Python, PyTorch, model fine-tuning, ESM-2, Node2Vec, RNA-seq pipeline, BigMHC, HPC/Slurm, and Git.

Genomic Analysis of Immunoglobulin Loci in Vertebrates 08/2022-07/2023
Dr. Yana Safonova at Johns Hopkins University *Baltimore*

· Performed comparative genomic analyses of immunoglobulin loci across mammalian species to investigate the role of transposable elements in evolutionary diversification.
· Contributed to phylogenetic tree construction and figure development for an unpublished study comparing the immunoglobulin locus of Peromyscus leucopus (white-footed mouse) with related species.
· Worked to identify immunoglobulin subgroups and corresponding gene subsets across selected species using classification methods from the international ImMunoGeneTics Information System (IMGT).
· Tools and techniques: IgDetective, RepeatMasker, Python, Iroki, dot plot alignment, and clustering analysis.

In Silico Analysis of the Effects of KRAS Mutations 01/2022-05/2022
Dr. Xianghua Li at Zhejiang-Edinburgh Institute *Remote*

· Investigated the impact of all possible KRAS mutations on KRAS–KRAS dimerization affinity by computing $\Delta\Delta G$ of folding and binding in both heterozygous and homozygous forms (PDB: 6BOF).
· Identified allosteric residues beyond the binding interface with potential effects on KRAS dimerization.
· Compared the effects of oncogenic and non-oncogenic KRAS mutations on protein-protein binding affinity.

- Tools: FoldX, FreeSASA, R, Python, and shell scripting.
- **Honours Dissertation:** *In Silico Analysis of Mutational Effects of KRAS on its Dimerization as Heterozygotes and as Homozygotes* (unpublished; available at bit.ly/AdeleHonoursDissertation).

Computational Modeling of Bacteriophage Adsorption to Bacteria 08/2020-09/2021
Dr. Sebastian Leptihn and Dr. Melanie Stefan at Zhejiang-Edinburgh Institute *Remote*

- Led a team of three students in developing rule-based and spatial stochastic models to investigate how receptor/co-receptor interactions and host cell polysaccharide depolymerization influence infection rates.
- Tools: BioNetGen Language, MCell, and CellBlender.

RNA-Seq Data Analysis of Primate Embryonic Stem Cells 06/2019-10/2019
Dr. Wanlu Liu at Zhejiang-Edinburgh Institute *Haining*

- Developed an RNA-seq analysis pipeline to identify differentially expressed transposable elements in primate embryonic stem cells using several samples from GSE106245.
- Tools: SRA Toolkit, FastQC, MultiQC, Cutadapt, HISAT2, Samtools, TEtranscripts, deepTools, and R/DESeq2.

LEADERSHIP EXPERIENCE

Indonesian Student Association at JHU 08/2023-12/2024
President *Baltimore*

- Founded the organization, recruited co-founders, co-authored its constitution, and secured official recognition.

Graduate Association of Women in CS and ECE (GRACE) 09/2023-09/2024
Professional Development Co-Chair *Baltimore*

- Coordinated a resume workshop with PHutures for 20 registered participants.
- Co-organized a mentoring night connecting undergraduate and graduate students.

Academic Family 09/2019-08/2020
Peer Mentor *Haining*

- Mentored eight students on academic and personal challenges, including language barriers, time management, and exam preparation.

Student-Staff Liaison Committee 09/2018-09/2020
Student Representative *Haining*

- Elected for two consecutive years to represent the cohort, conducted student surveys, and presented recommendations to university staff.

Wesley School for Seniors 07/2017-04/2018
Tech Tutor *Sydney*

- Led weekly digital literacy sessions for eight older adults.

NASA International Space Apps Challenge 01/2017-06/2017
Hackathon Co-Organizer *Sydney*

- Coordinated communications with mentors, sponsors, and 278 registered participants.
- Supported event marketing and volunteer coordination.

HONORS AND AWARDS

Fully funded throughout graduate studies at Johns Hopkins through Research Assistantships. 2022-2026
Awardee of the Zhejiang-Edinburgh Institute full-ride scholarship for the full duration of study. 2018-2022
Member of winning team in UK-China Joint Institute Alliance case-study challenge on student engagement. 2019